

GAIRA V7

The Canonical Inference Engine

Phase 09 — one frozen path from a Raman spectrum to an explainable biochemical interpretation

Status	COMPLETE — packaging phase, architecture unchanged
Engine	50 LSMs · 49 CSMs · 154 molecules · 16 chemistry axes
Atlas fingerprint	2e43ddcca7d3be41c5f9da016fb8277f
Deterministic	True
LSM reconstruction EV	0.8237 replicate consistency 0.8799
CSM chemistry top-1 / top-3	0.8453 / 0.9707 macro-F1 0.8068
Molecule top-1 / top-5 / MRR	0.6053 / 0.7947 / 0.6870
Chemistry top-1 (HELD OUT)	0.8507 top-3 0.9760 macro-F1 0.8110
Chemistry top-1 (in-sample)	0.9547 — a sanity check, NOT a performance claim
Chemistry calibration ECE	0.0534
Radar reproducibility	0.9596
Under perturbation	molecule 0.811 · chemistry 0.889 · radar cosine 0.965
Frozen baseline reproduced	True — exactly, all six retrieval metrics
Gates	16 of 16 PASS
Scope	pure Raman reference spectra only
Completed	2026-08-07 05:32:15 UTC

The Chemistry Evidence radar shows RELATIVE BIOCHEMICAL EVIDENCE. It is not a concentration, not an abundance, and not a mixture statement.
Sources of record: PHASE_09_REPORT.md · PHASE_09_ENGINE_SPEC.md · PHASE_09_MATHEMATICAL_APPENDIX.md · PHASE_09_SCIENTIFIC_AUDIT.md · PHASE_09_DECISION_GATE.md

Contents

This document is written to be read start to finish by someone who is not a spectroscopist. Every technical section is followed by a plain-English gloss marked in blue. Nothing in it is asserted without a number behind it, and the numbers all come from the committed artifacts of this phase.

Part I — What the engine is

- The problem, stated plainly
- The five stages
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Part II — The sixteen figures

Each figure is preceded by a page explaining what it shows, how to read it, and what it does not license you to conclude.

Part III — Results

- Validation 1-4 in full
- Per-class behaviour
- Robustness
- Gates

Part IV — Mathematics

Every equation the engine executes, each with its intuition.

Part V — Limits, decision and glossary



What the engine is

A Raman spectrum goes in. A ranked list of candidate molecules, a sixteen-dimensional chemistry reading, a calibrated confidence and a complete audit trail come out. One path, no alternatives, no tunable knobs.

The problem, stated plainly

Shine a laser at a sample and a small fraction of the light comes back with its colour shifted. The size of each shift is set by how the molecules in the sample vibrate, so the pattern of shifts is a fingerprint of the chemistry present. That pattern is a Raman spectrum.

Why this is hard

A textbook treats a spectrum as a fingerprint to be matched against a library. Real samples do not cooperate. Three facts govern everything in this architecture:

- Spectra are mixtures, not fingerprints. What arrives at the detector is the sum of contributions from everything in the illuminated volume.
- A peak is not a molecule. A band near 1450 cm⁻¹ says a CH₂ group is bending. Thousands of biological molecules contain CH₂ groups.
- Nearby is not the same. Published assignments frequently match a peak to a molecule because the wavenumbers agree to within a few units and the biology is plausible. That is not evidence.

What GAIRA answers instead

Rather than asking 'which molecule is this?', the engine answers a question it can support: what chemistry does the evidence favour, how strongly, and by way of which specific spectral features? It still returns ranked molecular candidates, because a shortlist is useful, but the load-bearing output is the chemistry reading.

In plain terms

Think of a doctor reading a blood panel. They rarely conclude 'this is precisely molecule X at precisely concentration Y'. They read a pattern and say which processes it is consistent with, and how sure they are. This engine is built to do the second thing well rather than the first thing badly.

The five stages

Five stages, always in this order, with no branches and no options.

1 • Preprocess

Crop to 450-1800 cm⁻¹, resample to 676 points spaced 2.0 cm⁻¹ apart, subtract the fluorescence background by asymmetric least squares, smooth with a Savitzky-Golay filter, and scale to unit length. This makes any two spectra comparable and discards the absolute intensity scale.

2 • Project onto 50 Local Spectral Motifs

Express the spectrum as a non-negative sum of 50 learned basis spectra. About ten are active for a typical sample. Reported for interpretability; not consumed by any later stage.

3 • Project onto 49 Consensus Spectral Motifs

The same operation against the merged, deduplicated dictionary. These 49 numbers are the canonical representation - the coordinates every downstream layer reads, and the only ones.

4 • Retrieve molecules

Compare the activation pattern against 154 reference molecules by cosine similarity, and rank them. Because a cosine is an inner product, the score decomposes exactly into per-motif contributions, which is what makes the ranking explainable rather than merely produced.

5 • Aggregate to Chemistry Evidence

Collapse the activation into 16 numbers, one per chemistry family, using a hierarchical model that lets coarse chemistry gently inform fine chemistry without ever excluding it. Calibrate, and emit the radar with its confidence and warnings.

In plain terms

Each stage answers a narrower question than the last: is this spectrum clean, what patterns is it made of, which of those patterns are real and distinct, what does it resemble, and what does that mean chemically.

What is deliberately absent, and why

The most informative part of this architecture is what it does not contain. Each of the following was built, measured, and rejected on evidence.

excluded	measured outcome	phase
Theme layer	chemistry on unseen molecules fell to 0.405	03 / 04
Meta Components	0.392	04.5
11 grounded axes	0.664	04
Latent geometry / coordinates	molecule +0.016, McNemar p = 0.180	06.5
Chemistry-aware reranking	no gain once both arms used the same bank	08
BSV2 on the inference path	derived description, never an input	07
SERS / serum / EV handling	out of scope; unmeasured in V7	—

For comparison, chemistry accuracy on unseen molecules using the raw preprocessed spectrum is 0.608, and using the CSM activation it is 0.855. Every layer built ABOVE the CSM performed worse than the CSM itself.

The failure mode this rebuild kept catching

Seven times across V7, a selection rule was maximised by an answer that was degenerate rather than good: a calibrator that returned the base rate for every input scored best on calibration error; a cluster count of 4 scored best on bootstrap stability because a coarse partition is trivially stable; a programme count of 16 over a 16-dimensional input scored best because the model had learned the identity function.

The countermeasure that worked every time was the same: describe what a degenerate answer would look like BEFORE running the selection, and add a floor that disqualifies it.

In plain terms

A metric is a proxy for what you want. Any proxy can be satisfied by something you did not want. The discipline is not cleverer metrics — it is naming the cheat in advance and forbidding it.

How to read this document

Every figure in Part II is preceded by an explanation page laid out the same way.

Title and opening line	what the figure is for, in one sentence
Three or four paragraphs	what each panel shows and how to read it
A blue box	the plain-English gloss, or the caveat that matters most
The figure itself	full page, with its formal caption beneath

Conventions used throughout

- Wavenumbers are in cm^{-1} and always span 450 to 1800.
- 'Held out' means molecule-grouped cross-validation: every spectrum of a molecule sits in one fold, so the model is always predicting a molecule it has never seen. This is the honest number.
- 'In-sample' means the model was fitted on all 375 spectra and asked about those same spectra. It describes the object being shipped and is never a performance claim. Where both appear, quote the held-out one.
- 'Top-k' is the fraction of queries whose correct answer appeared in the first k results. 'Macro F1' averages over classes so a 7-spectrum class counts as much as an 80-spectrum one.
- Radar radius is relative evidence. Never a concentration.

Where to start

If you read only one figure, read Figure 16. If you read two, add Figure 14.



The sixteen figures

Each figure gets an explanation page and a full page of its own. Figures 1 to 6 describe how the engine was built; 7 to 9 show it working; 10 to 13 measure it; 14 to 16 stress it and summarise.

FIGURE 1

The map of everything

Read the diagram top to bottom. Everything above the horizontal rule happened once, during Phases 00-06, and is now frozen: the corpus was assembled, the motif dictionaries were learned, the chemistry map was fitted. Everything below the rule happens every time a spectrum arrives.

A spectrum enters at the bottom left and moves right through five stages. It is cleaned, described as a mixture of 50 local motifs, re-described as a mixture of 49 consensus motifs, compared against 154 reference molecules, and summarised as evidence along 16 chemistry axes.

The red panel is as important as the blue one. Each item in it was built, measured and rejected: themes lost information, Meta Components lost more, the geometric coordinate layer produced an improvement that a significance test could not distinguish from zero. They are absent because of evidence, not taste.

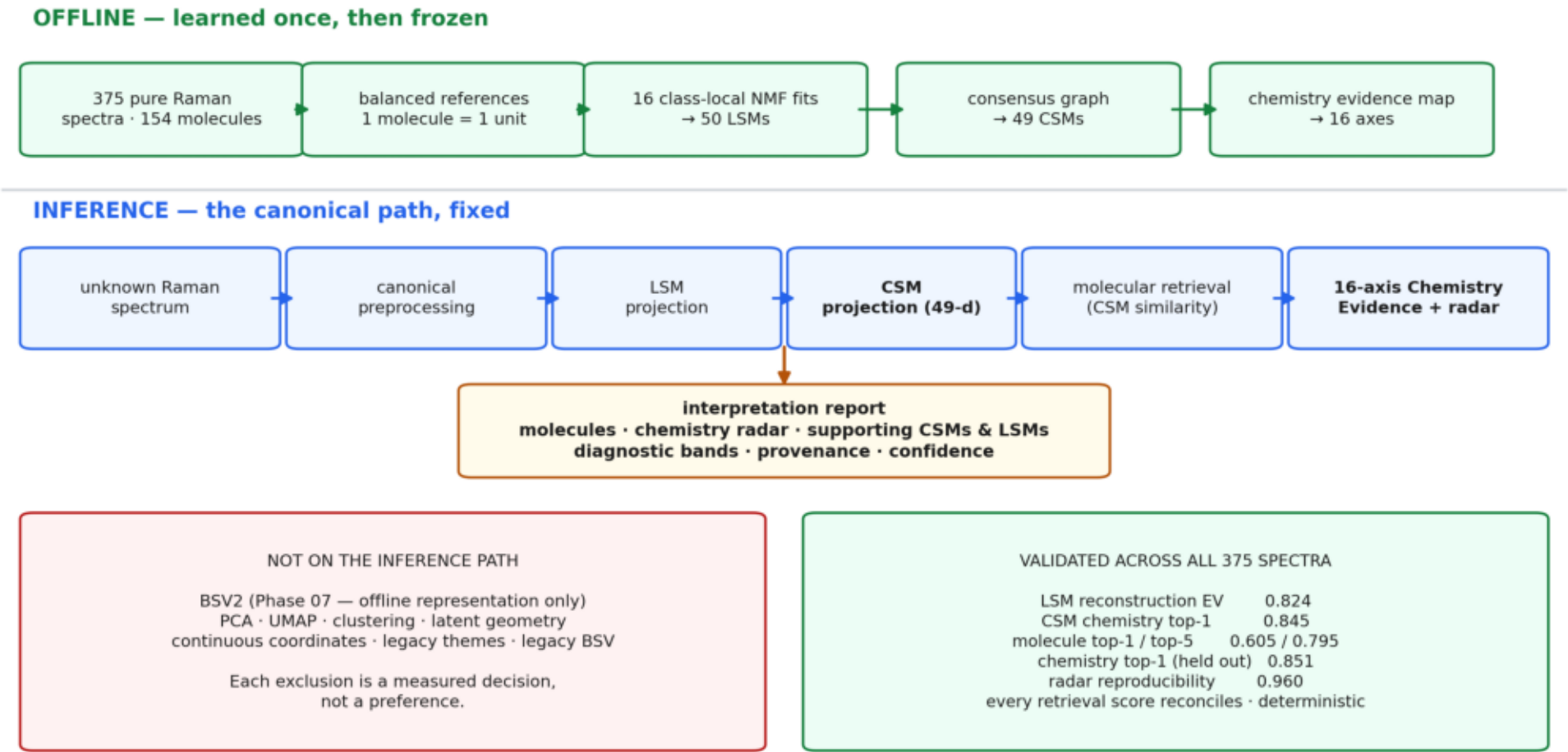
BSV2 sits deliberately off to the side. It is a way of describing the chemistry output for a human reader; it never feeds back into the answer.

In plain terms

A diagram like this is usually aspirational. This one is enforced: the engine verifies four cryptographic fingerprints when it loads and refuses to start if any frozen component has changed.

Figure 01

Figure 1 · The complete GAIRA V7 architecture — offline learning above, frozen inference below



The complete GAIRA V7 architecture. Above the rule is offline learning, done once and then frozen; below it is the canonical inference path fixed by Phases 05-08. The red panel lists what is deliberately absent, and every exclusion is a measured decision.

FIGURE 2

What the engine was built from

Every claim in this document rests on 375 spectra of 154 pure molecules. Knowing the shape of that corpus is the fastest way to understand both what the engine can do and where it must stop.

The class sizes are very uneven: 80 peptide/protein spectra against 3 nucleic-acid polymer spectra. That imbalance is why performance is reported as macro F1, which gives every class one vote, alongside plain accuracy.

The replicate histogram is the panel to dwell on. 66 of 154 molecules appear exactly once. When such a molecule's only spectrum is held out for testing, the correct answer is not in the reference bank at all, and the engine cannot possibly retrieve it. That accounts for 68 of the 375 test queries.

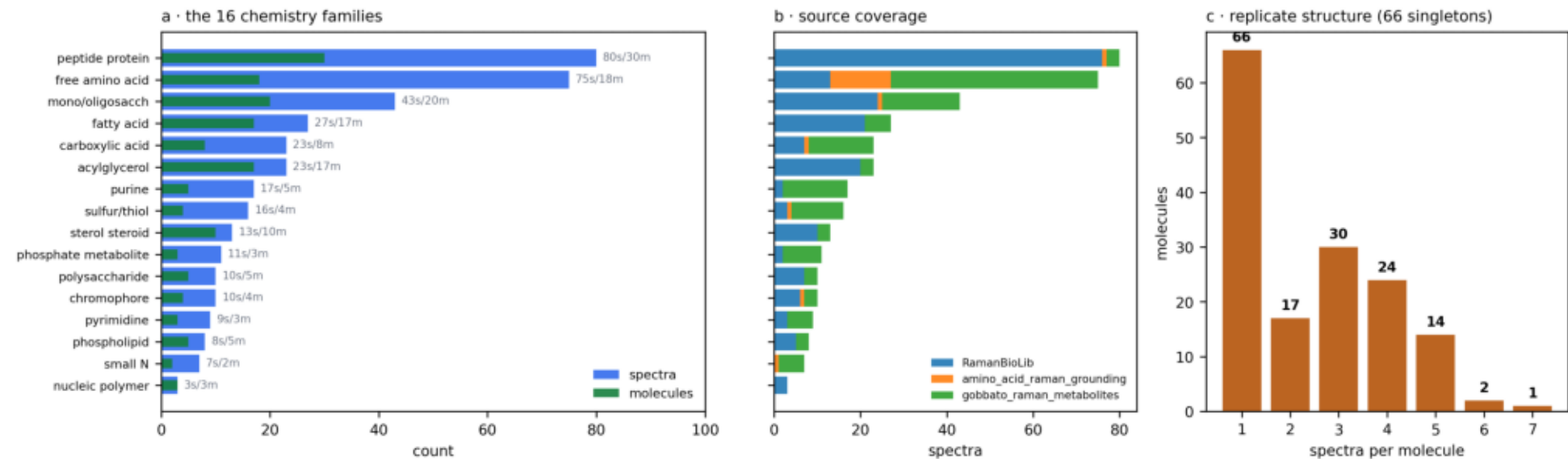
Three source libraries contribute. Phase 06.5 measured how much of the spectral geometry is explained by chemistry versus by which library a spectrum came from: chemistry wins by a factor of eleven, but the acquisition signal is not zero.

In plain terms

This is a corpus of PURE compounds measured under clean conditions. It contains no serum, no extracellular vesicles, no SERS substrates and no tissue. Nothing measured here transfers to those settings without being measured again.

Figure 02

Figure 2 · The grounded Raman corpus — 375 spectra, 154 canonical molecules, 16 chemistry families



The grounded Raman corpus: 375 spectra, 154 canonical molecules, 16 chemistry families, three source libraries. 66 molecules have a single spectrum, which sets a hard ceiling on molecular retrieval.

Making spectra comparable

Two instruments measuring the same molecule produce different numbers: different wavenumber points, different intensity scales, different fluorescent backgrounds. Preprocessing removes those differences and keeps what is chemical.

Resampling puts every spectrum on the same 676-point ruler from 450 to 1800 cm^{-1} . Where a spectrum does not reach, the engine writes zeros and says so in a warning. It never extrapolates, because an invented value would be indistinguishable from a measured one further downstream.

The baseline stage is the subtle one. Fluorescence adds a broad hump that can dwarf the peaks. Asymmetric least squares fits a smooth curve that is pulled hard by the valleys and barely at all by the peaks, so it traces the background rather than the signal. The asymmetry is the entire trick.

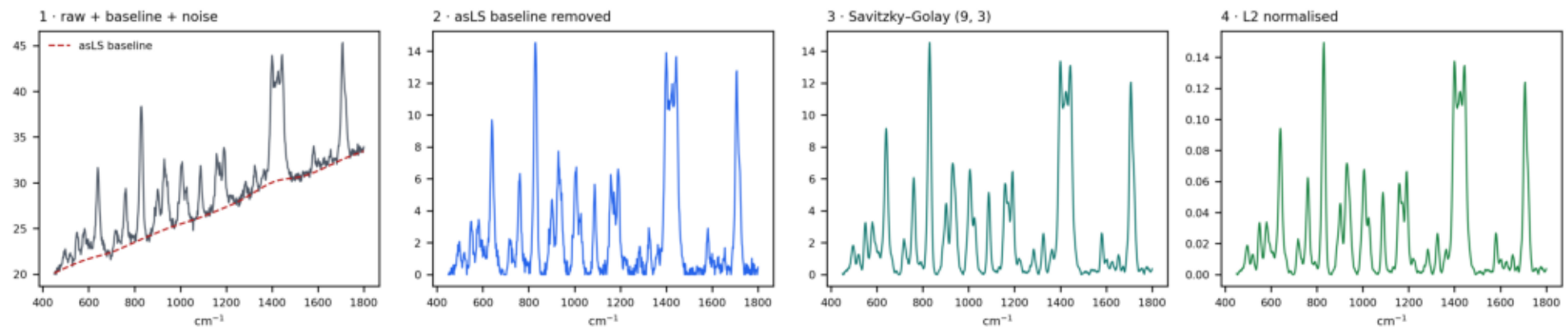
Savitzky-Golay smoothing fits a small cubic in a sliding window instead of taking a moving average, so it removes jitter without flattening peaks. Finally the spectrum is scaled to unit length, which discards laser power and integration time.

In plain terms

That last step is why no output of this engine can be read as a concentration. The absolute scale is deliberately destroyed before anything else happens.

Figure 03

Figure 3 · Canonical preprocessing — crop to 450-1800 cm^{-1} , resample to 676 bins at 2.0 cm^{-1} , asLS baseline, SG smoothing, L2 normalisation



Canonical preprocessing stage by stage: crop to 450-1800 cm^{-1} , resample to 676 bins at 2.0 cm^{-1} , remove the fluorescence baseline by asymmetric least squares, smooth with Savitzky-Golay, normalise to unit length. Unchanged since V5.

Learning the vocabulary

A Raman spectrum of a mixture is a sum of its components' spectra. Non-negative matrix factorisation takes that literally: it looks for a small set of basis spectra that can be ADDED, never subtracted, to reproduce what was observed.

Each Local Spectral Motif is one such basis spectrum. Fifty of them, learned separately inside each of the 16 chemistry families, so a motif describes a pattern that its family actually contains rather than an average across unrelated chemistry.

The non-negativity is a scientific constraint, not a technical convenience. Methods that allow negative weights - principal component analysis, for one - fit better but produce components with negative lobes, and there is no such thing as negative-two-parts glucose in a real mixture.

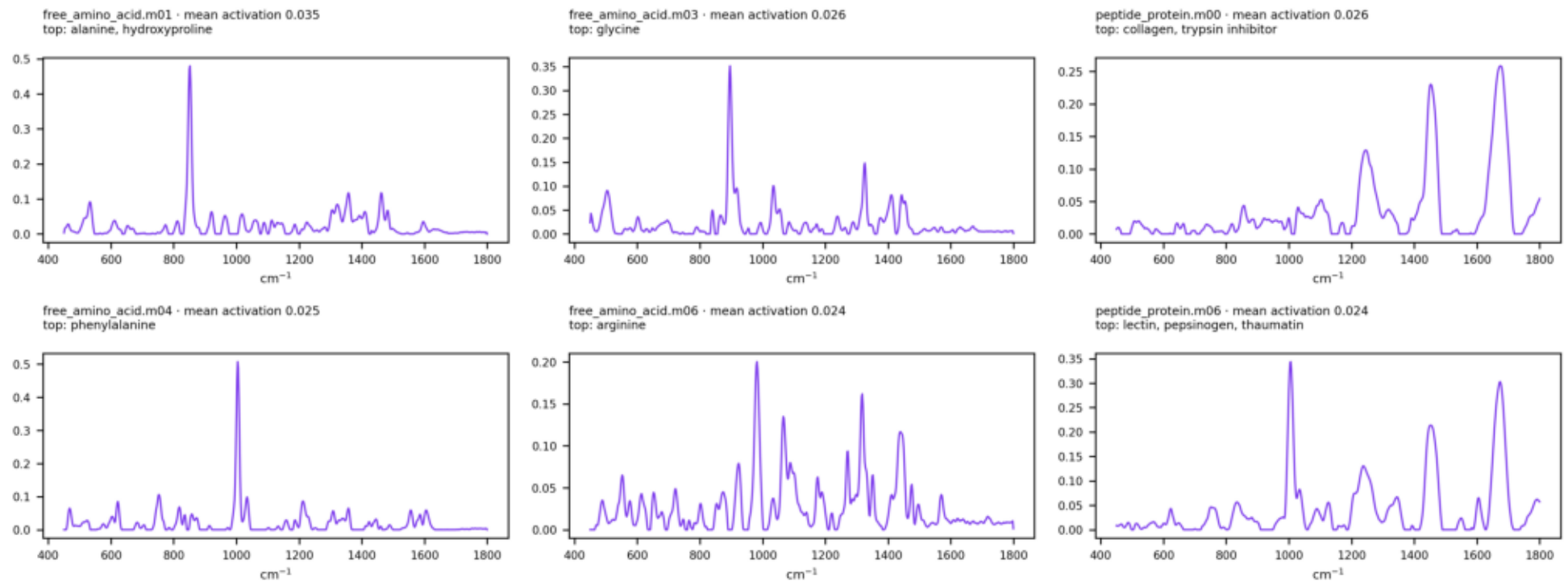
A typical spectrum uses about ten of the fifty motifs. The dictionary is describing composition, not memorising curves; if it were memorising, each spectrum would activate its own private motif and nothing would generalise.

In plain terms

Explained variance averages 0.824 across the corpus, but the minimum is 0.209. The molecules at the bottom - pyruvate, thymine, malic acid, urea - are small and high-symmetry, with few strong bands in this window. The engine reports low confidence on them rather than guessing.

Figure 04

Figure 4 · Representative Local Spectral Motifs — the six most-activated of 50



Six representative Local Spectral Motifs of 50. Each is a non-negative basis spectrum learned inside one chemistry class, so it describes a pattern that class actually contains rather than one imposed on it by the corpus average.

FIGURE 5

From 50 motifs to 49 coordinates

Some local motifs describe the same underlying vibration seen from two different chemistry families. Phase 02 merged those by building a graph in which motifs are nodes and an edge means 'these two are the same thing'.

An edge weight combines seven independent similarity measures using a weighted GEOMETRIC mean. This matters: under an ordinary average, two motifs that merely looked alike could be merged on the strength of one channel. Under a geometric mean, any single near-zero channel drags the whole weight toward zero. It is the operational form of 'a consensus requires several independent lines of evidence'.

The threshold for cutting the graph was not chosen by eye. It was swept from 0.05 to 0.90 and tested against a null built by shuffling spectral bands, keeping only groupings that survive a wide contiguous range of thresholds.

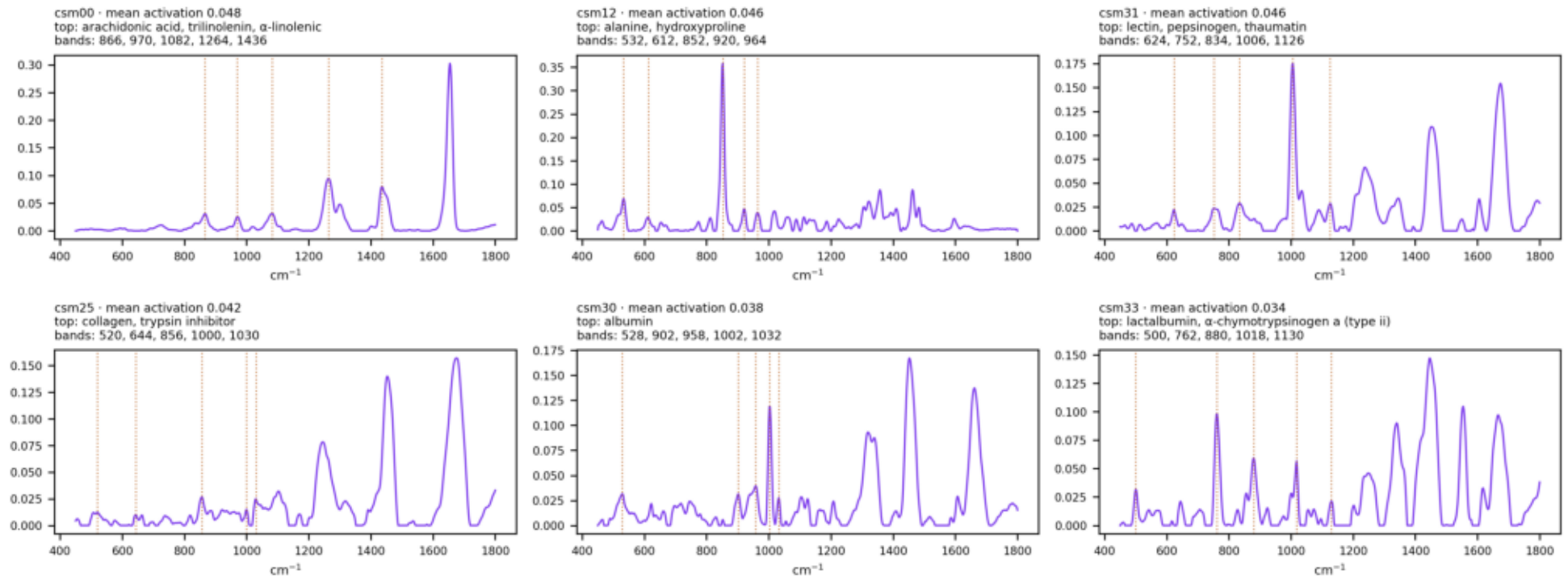
The result is 49 Consensus Spectral Motifs, each carrying its diagnostic band positions - the amber marks. Those bands are what lets a chemistry conclusion be traced all the way back to specific wavenumbers.

In plain terms

Compressing 50 motifs to 49 costs essentially nothing in reconstruction (0.8237 to 0.8232) while RAISING replicate consistency from 0.880 to 0.893. That is the whole reason the layer exists, and it is why the CSM activation - not the LSM activation - is the canonical representation.

Figure 05

Figure 5 · Representative Consensus Spectral Motifs — the six most-activated of 49 (amber = diagnostic bands)



Six representative Consensus Spectral Motifs of 49, diagnostic bands marked in amber. These 49 numbers are the canonical coordinates that every downstream layer reads.

FIGURE 6

The shape of chemistry space

This is a two-dimensional shadow of a 49-dimensional space, produced by UMAP. It is here to build intuition and for no other purpose.

Molecules of the same chemistry family land near each other without ever having been told to. Phase 06.5 quantified this properly: chemistry explains 45.2% of the distance structure, excitation wavelength 11.8%, and source library 4.0%. Chemistry wins by a wide margin, but acquisition is not nothing.

Phase 06.5 also asked whether the space contains a natural number of clusters. It does not. Across four algorithms and seven internal indices, not one index has an interior optimum - they rise or fall monotonically as the cluster count increases. The space is a continuum with dense regions, not a set of discrete islands.

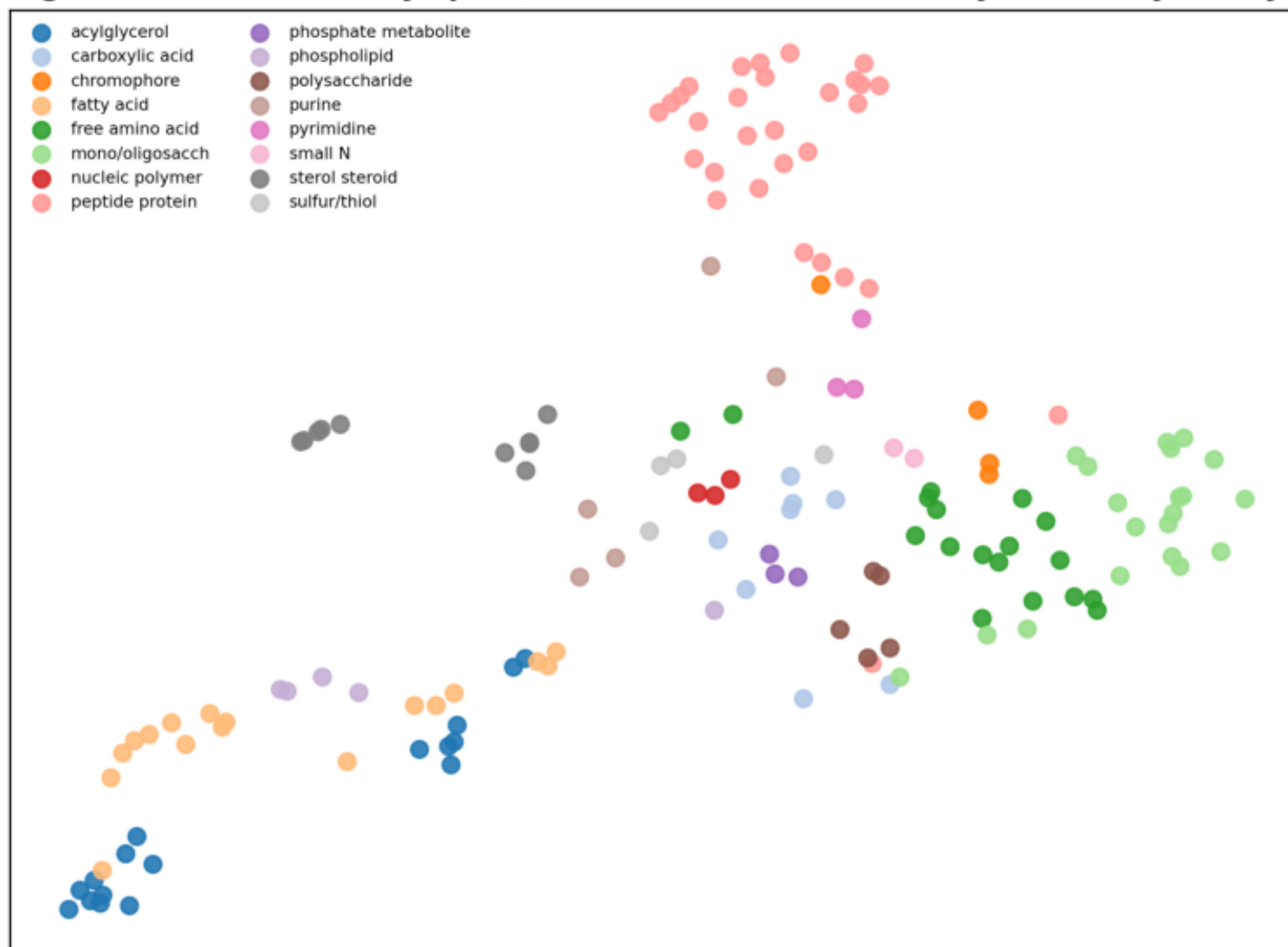
That result has a consequence worth stating: the 16 chemistry classes are a curated cut through a continuum, chosen because chemists find them useful. They are not a discovery, and the geometry does not validate them.

In plain terms

Do not read distances on this plot quantitatively. UMAP preserves local neighbourhood structure and distorts global distance freely; two clusters appearing far apart here may not be far apart in the real space.

Figure 06

Figure 6 · CSM similarity space, 154 molecules coloured by chemistry family



VISUALISATION ONLY — UMAP is not used during inference. Distances in this plot are not quantitative.

The CSM similarity space, one point per molecule, coloured by chemistry family. VISUALISATION ONLY - UMAP is not used during inference and the distances shown here are not quantitative.

FIGURE 7

One complete answer

This is the whole engine on one page, for one spectrum. Six panels, in the order the engine produces them.

Top left is the preprocessed spectrum. Next to it, the CSM activation: which of the 49 consensus motifs the spectrum is made of. About ten are active; the rest are exactly zero.

The retrieval panel ranks the 154 reference molecules by how closely their activation pattern points in the same direction as this one. Crucially, that similarity is an inner product, so it decomposes EXACTLY into per-motif contributions. The engine checks this for every candidate it reports: the parts sum to the whole to within $1e-9$. There is no hidden term.

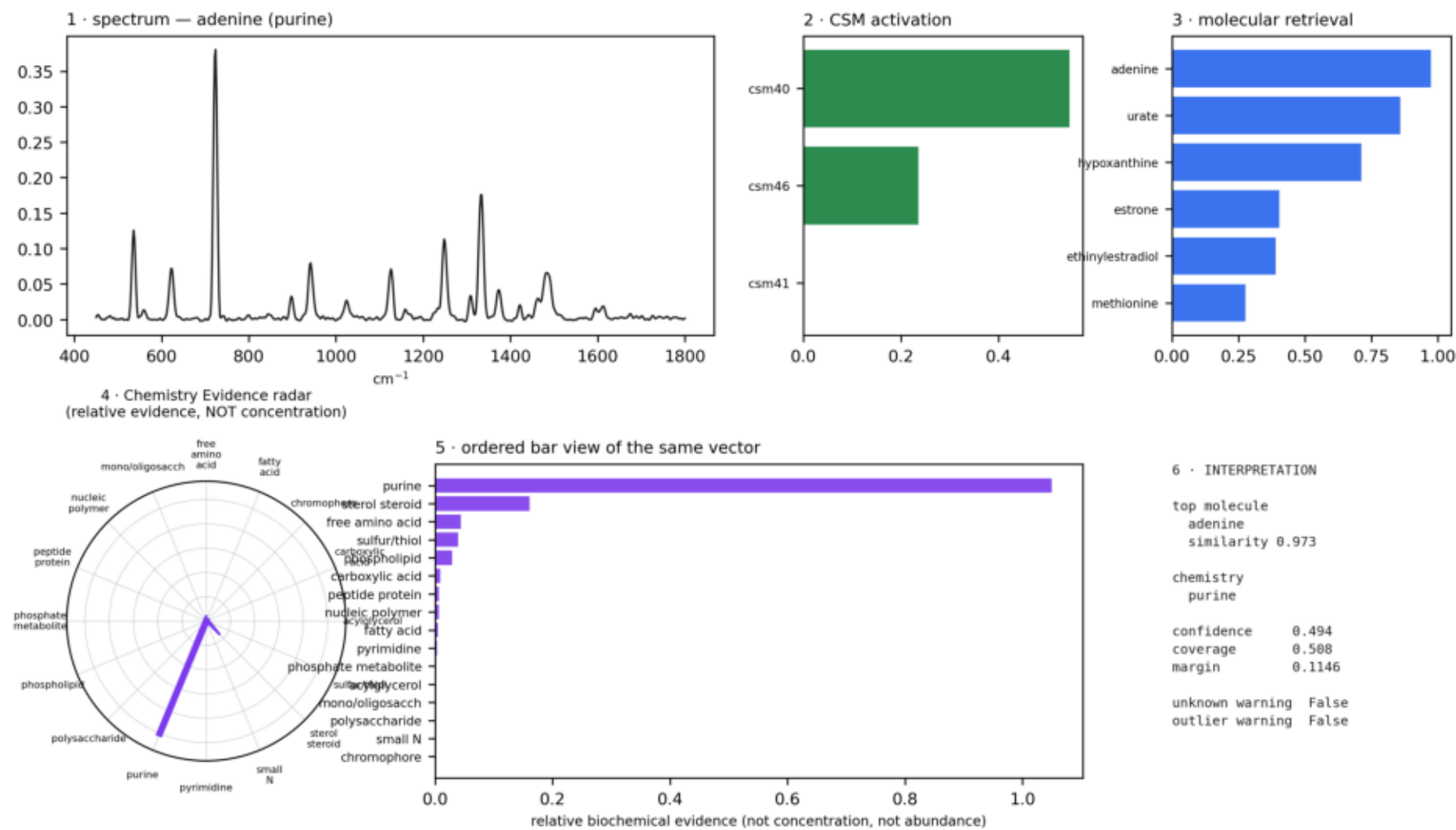
The radar and the bar chart are two views of the same 16 numbers. The radar is good for recognising a pattern at a glance; the bars are better for reading off an ordering. The written interpretation at the bottom right states the confidence and any warnings.

In plain terms

Confidence is the product of two things: how much of the spectrum the atlas could explain, and how well the best reference matched. Both must be high. A spectrum the dictionary cannot express might still land near some molecule by accident, and multiplying ensures that accident cannot become a confident answer.

Figure 07

Figure 7 · One inference, end to end



A single inference end to end: spectrum, CSM activation, molecular retrieval, chemistry radar, ordered bar view, and the written interpretation with its confidence and warnings. This is exactly what the engine returns.

Reading a radar

Sixteen spokes, one per chemistry family, always in the same order. The distance along a spoke is the relative evidence for that chemistry. Thicker spokes mean the engine is more confident.

Compare the shapes, not the sizes. A sterol produces a radar with one dominant spoke; an amino acid produces a broader shape with weight spread over several nitrogen-bearing families. It is the pattern that identifies the chemistry.

The same molecule measured twice produces nearly the same radar - cosine similarity 0.960 across replicates. That reproducibility is what makes the radar usable as a report to a human rather than a curiosity.

Under noise the radar is the last thing to break. Averaged over 35 perturbation conditions it retains cosine 0.965 to its clean self, while molecule identity falls to 0.811. The general answer survives longer than the specific one, which is exactly the intended behaviour.

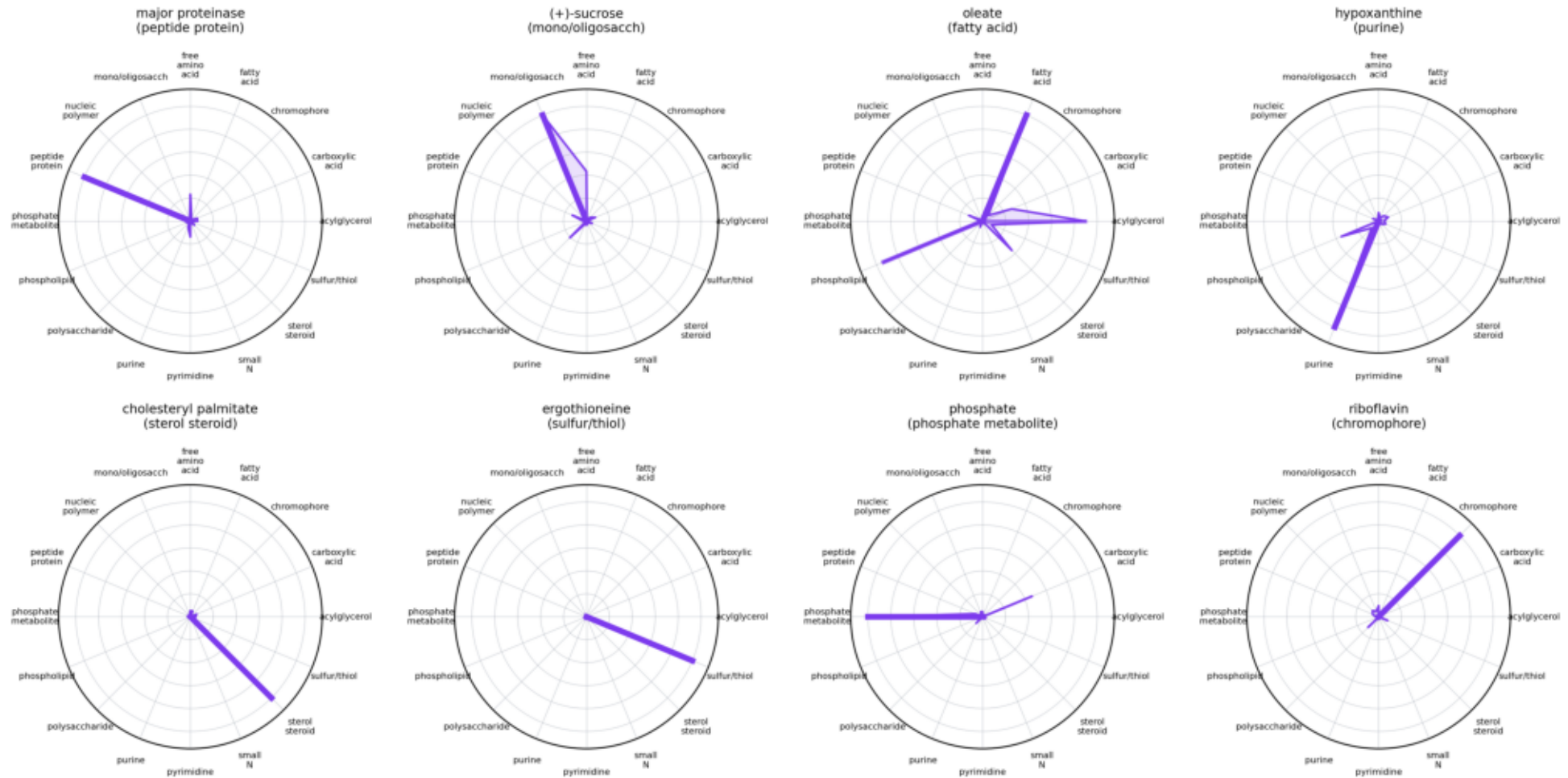
In plain terms

The radius is RELATIVE BIOCHEMICAL EVIDENCE. It is not a concentration, not an abundance, and not a mixture fraction. The absolute scale was destroyed by normalisation in the very first stage, and a similarity is not a quantity of material. A tall spoke means 'evidence associated with this chemistry is present', and nothing more.

Figure 08

Figure 8 · Chemistry Evidence radars across eight chemistry families

spoke thickness encodes calibrated confidence · radius is RELATIVE BIOCHEMICAL EVIDENCE, not concentration and not abundance



Chemistry Evidence radars across eight families. Spoke thickness encodes calibrated confidence. The radius is RELATIVE BIOCHEMICAL EVIDENCE - not a concentration, not an abundance, not a mixture fraction.

FIGURE 9

When retrieval works and when it does not

Three real queries, chosen to show the range. Green marks the true molecule wherever it appears in the ranking.

The left panel is the common case: the correct molecule is first, with a clear gap to the second. This happens for 60.5% of spectra.

The middle panel is a near miss - the true molecule is second or third, usually behind a close structural relative. A further 15.7% of spectra land here, which is why top-3 reaches 0.763. For a chemist, a shortlist of three is often more useful than a single confident guess.

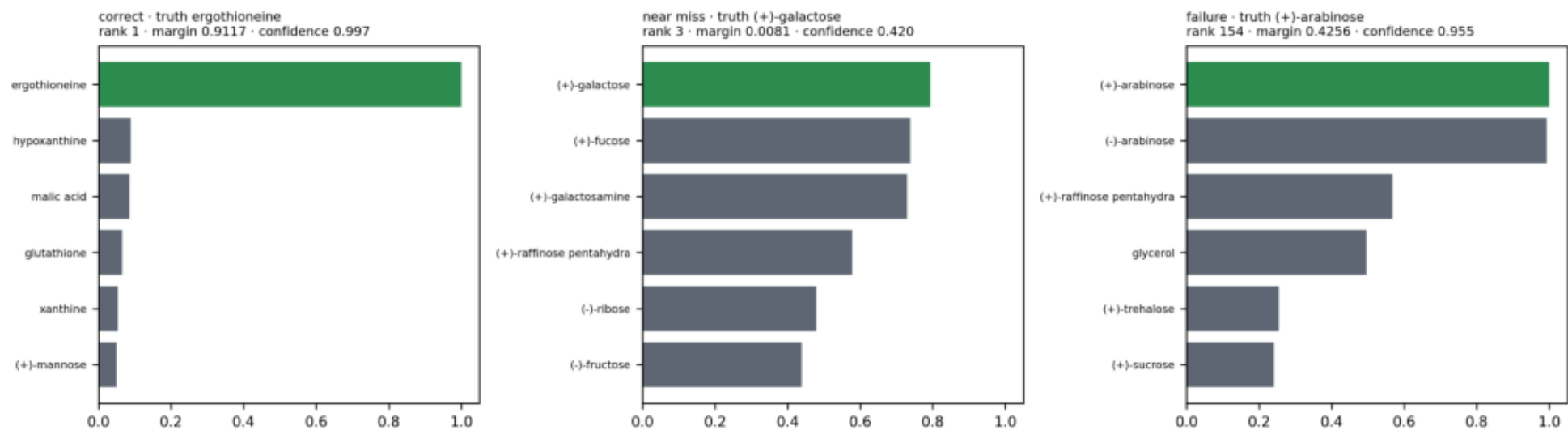
The right panel is a genuine failure. Many of these are structural rather than algorithmic: 68 of the 375 test queries belong to molecules with only one spectrum in the corpus, so when that spectrum is held out the answer is simply absent from the bank.

In plain terms

Those 68 unretrievable spectra are counted as failures. Excluding them would raise the headline from 0.605 to roughly 0.74, and that number would be misleading - the engine really did fail to name them.

Figure 09

Figure 9 · Retrieval examples — green is the true molecule



Three retrieval outcomes: a confident correct identification, a near miss where the true molecule sits second or third, and a failure. Green marks the true molecule.

FIGURE 10

Where the chemistry layer gets confused

Rows are the true class, columns the predicted one. A perfect result would put every spectrum on the diagonal.

The off-diagonal mass is not random. Fatty acids are confused with acylglycerols; free amino acids with peptides; purines with pyrimidines. In each case the two families share the chemical group that dominates their Raman signature - a long aliphatic chain, an amide backbone, a nitrogen heterocycle.

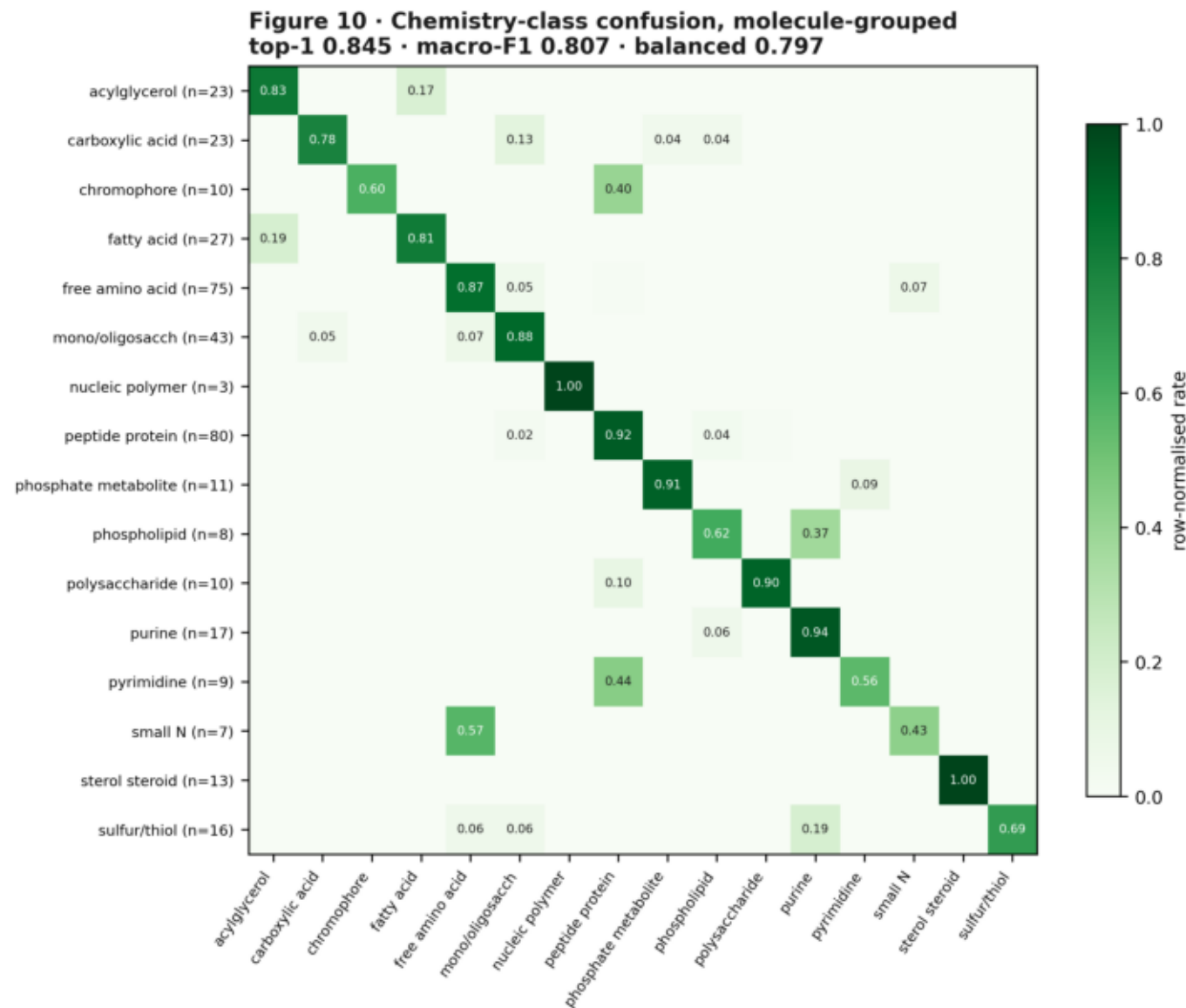
The weakest classes are the smallest. `small_nitrogenous` (7 spectra, F1 0.400) and `phospholipid_sphingolipid` (8 spectra, F1 0.556) are both dominated by larger families they resemble. `sterol_steroid` and `nucleic_acid_polymer` reach F1 1.000.

The ontology was deliberately NOT modified in response to this figure. Merging two classes because a model confuses them would improve the score while destroying the distinction the score was meant to measure.

In plain terms

This matrix is computed under molecule-grouped cross-validation: every spectrum of a molecule sits in the same fold, so the model is always predicting a molecule it has never seen. That is the honest question and it produces the honest 0.851.

Figure 10



Chemistry-class confusion under molecule-grouped cross-validation. The errors sit where a spectroscopist would expect them: small classes dominated by the chains and rings they share with larger ones.

FIGURE 11

How separable each chemistry is

A receiver operating characteristic curve asks: as we lower the bar for calling a spectrum 'this chemistry', how quickly do we pick up true cases relative to false ones? The area beneath is the chance that a random true case outranks a random false one.

Every axis sits near the top-left corner, and the macro-averaged area is 0.999. Taken at face value that would be an extraordinary result.

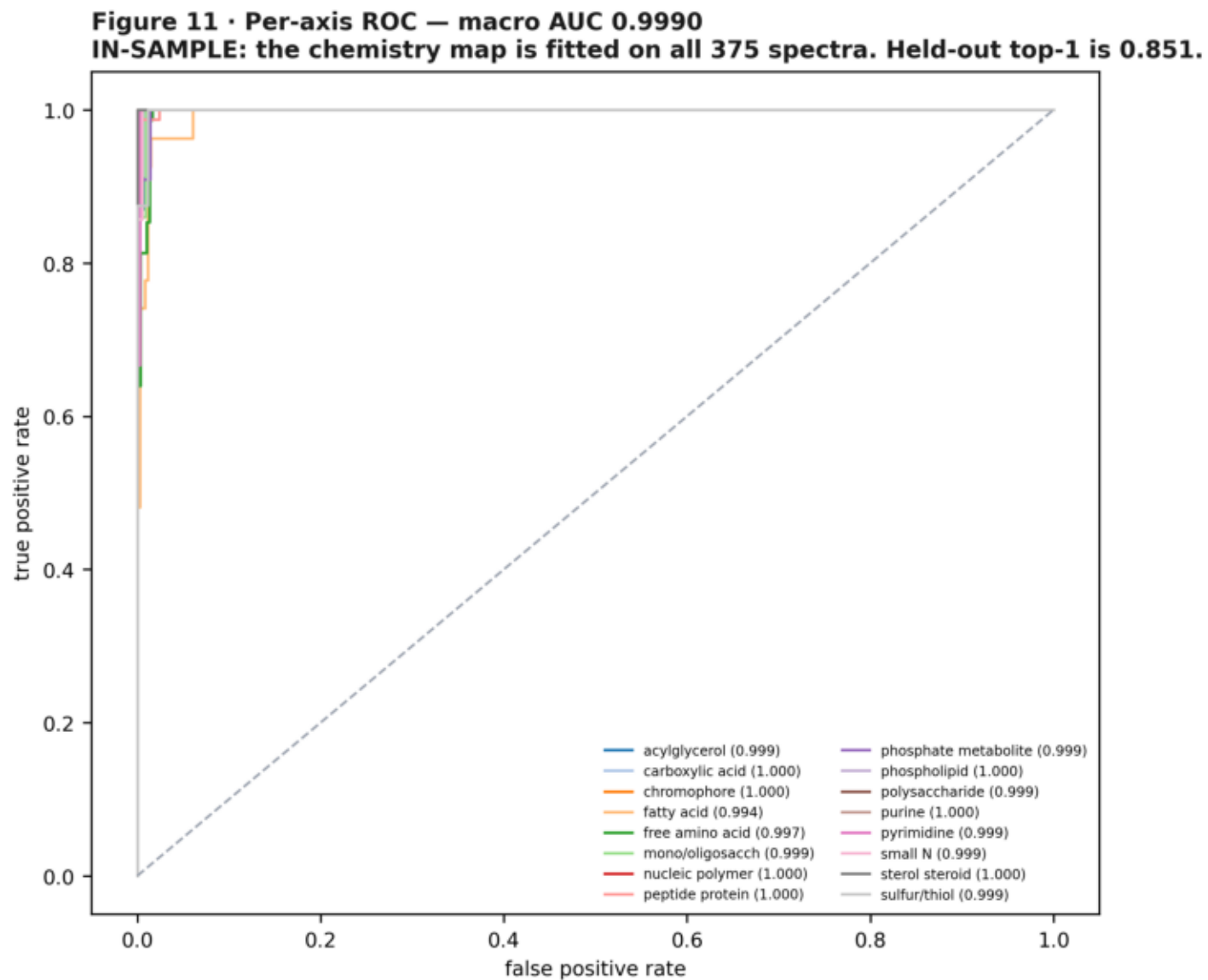
It should not be taken at face value. These curves are IN-SAMPLE: the chemistry map was fitted on all 375 spectra, and is being asked about those same spectra. That is the right way to describe the model being shipped, and the wrong way to predict performance on anything new.

The honest comparison is the held-out figure of 0.851 top-1 accuracy, computed with the model refitted inside each cross-validation fold. The gap between 0.955 in-sample and 0.851 held-out is precisely the size of the illusion.

In plain terms

This distinction was a defect in this phase before it was a caption. Validation 4 was originally computed in-sample only, and would have reported 0.955 as the headline. Gates G15 and G16 were added so it cannot recur.

Figure 11



Per-axis ROC. These curves are IN-SAMPLE - the chemistry map is fitted on all 375 spectra, as a shipped engine would be. The held-out top-1 of 0.851 is the number to quote as performance.

FIGURE 12

The same question, asked more strictly

Precision-recall curves plot how often the engine is right when it commits, against how many true cases it finds.

ROC has a blind spot under class imbalance. With 80 peptide spectra and 3 nucleic-acid-polymer spectra, a large pool of true negatives keeps the false positive rate small no matter how the model behaves on the rare class. Precision has no such escape hatch: a false positive hurts immediately.

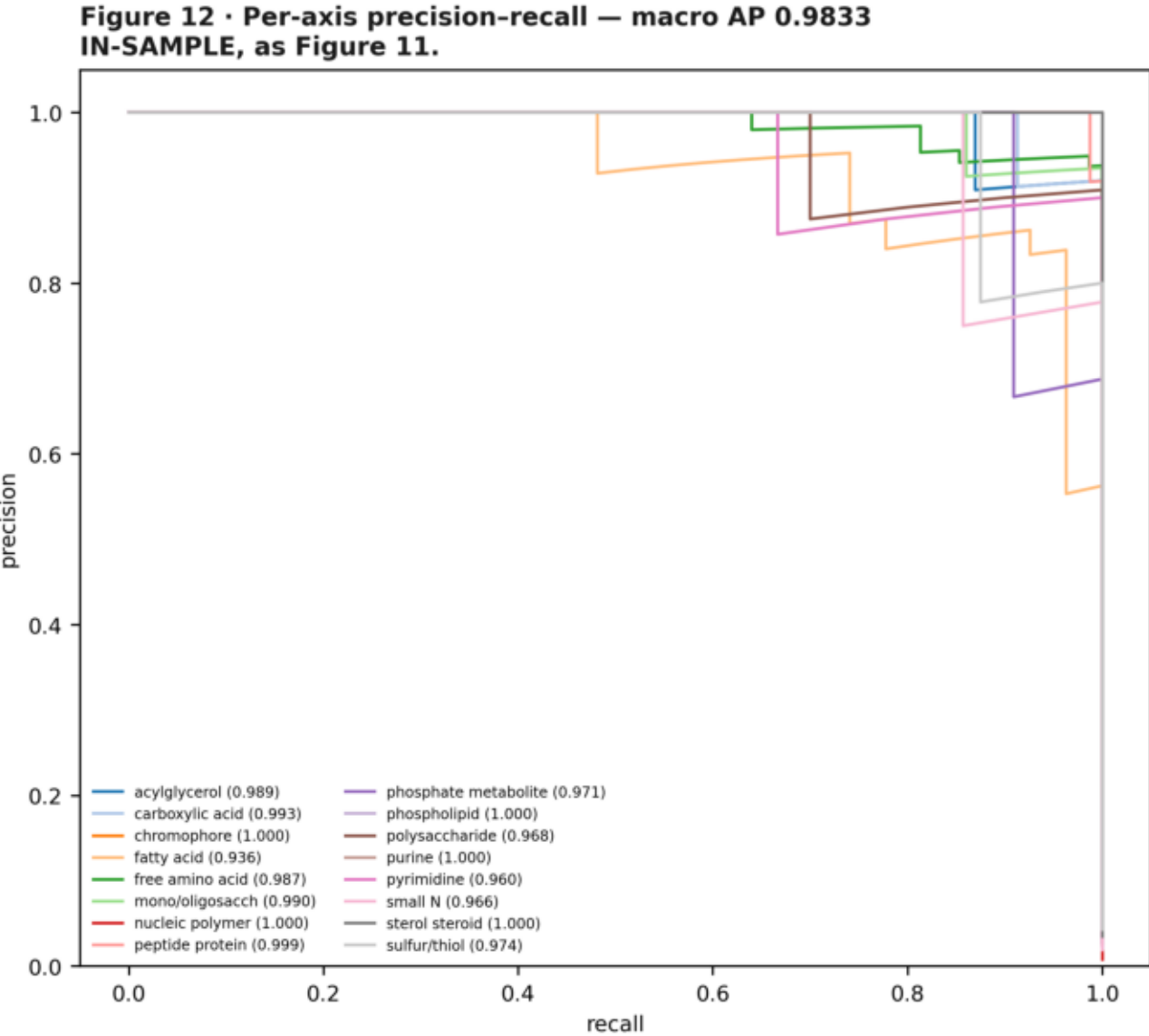
For an unevenly sampled corpus like this one, PR is therefore the more honest picture. Macro average precision is 0.983 - still high, still in-sample.

The classes whose PR curves sag are the same ones the confusion matrix flagged: the small families that share their strongest bands with larger neighbours.

In plain terms

Same in-sample caveat as Figure 11. Quote 0.851.

Figure 12



Per-axis precision-recall, with the same in-sample caveat as Figure 11. Under class imbalance PR is the more honest of the two curves.

FIGURE 13

Knowing when not to answer

Three panels about honesty rather than accuracy.

The reliability diagram asks: of everything the engine called 80% likely, was it right 80% of the time? Points on the diagonal mean the confidences can be believed. Expected calibration error summarises the gap in one number.

Calibration error alone is a trap, and V7 learned it the hard way. A model that outputs the same base rate for every input has near-perfect calibration and is completely useless. That is why model selection here uses log loss subject to FLOORS on sharpness and discrimination - a constant predictor is disqualified before calibration is consulted at all.

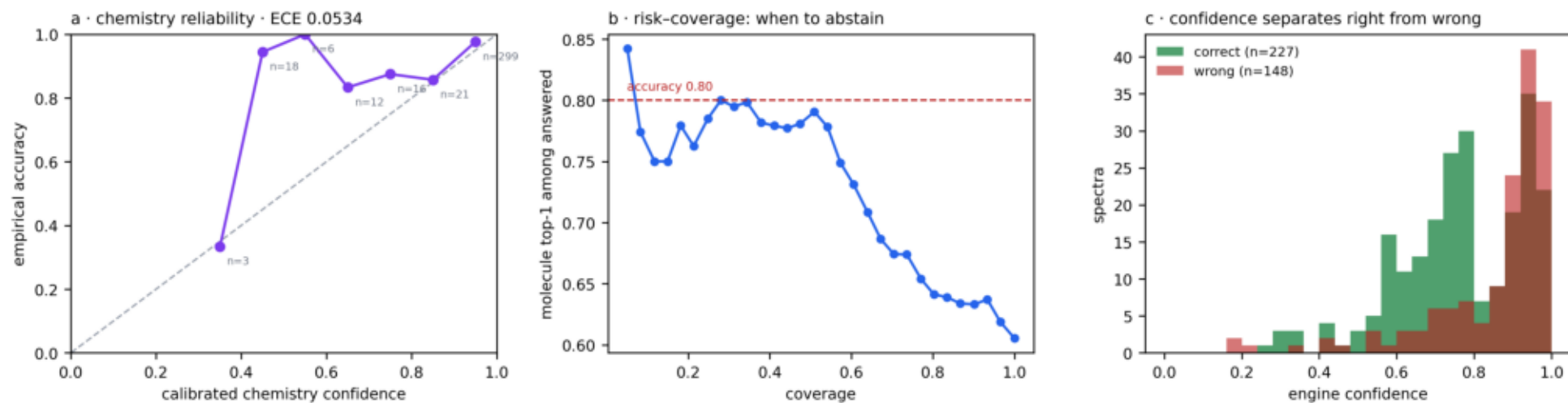
The risk-coverage curve is the practical one. It says: if you only answer when confidence exceeds some threshold, how often are you right, and how many spectra do you answer at all? Refusing to answer below a margin of 0.497 keeps 51% of spectra at 79% accuracy, against 61% at full coverage.

In plain terms

No operating point is baked into the engine. The curve is published so that an operator can choose one for their own tolerance for being wrong.

Figure 13

Figure 13 · Calibration and selective prediction



Calibration of the chemistry confidence, the risk-coverage curve that tells an operator when to abstain, and the separation between confident-correct and confident-wrong answers.

FIGURE 14

What happens as the signal degrades

Seven ways a spectrum degrades on a real instrument, each applied at five strengths, each run through the complete engine from raw spectrum to final radar.

Detector noise and shot noise barely register - the radar holds above cosine 0.998 at every level. Preprocessing absorbs high-frequency perturbations before the projection ever sees them.

Baseline drift and fluorescence are harder, because they attack the stage designed to remove them. At the most extreme drift the engine still classifies chemistry correctly 72% of the time and retains radar cosine 0.867.

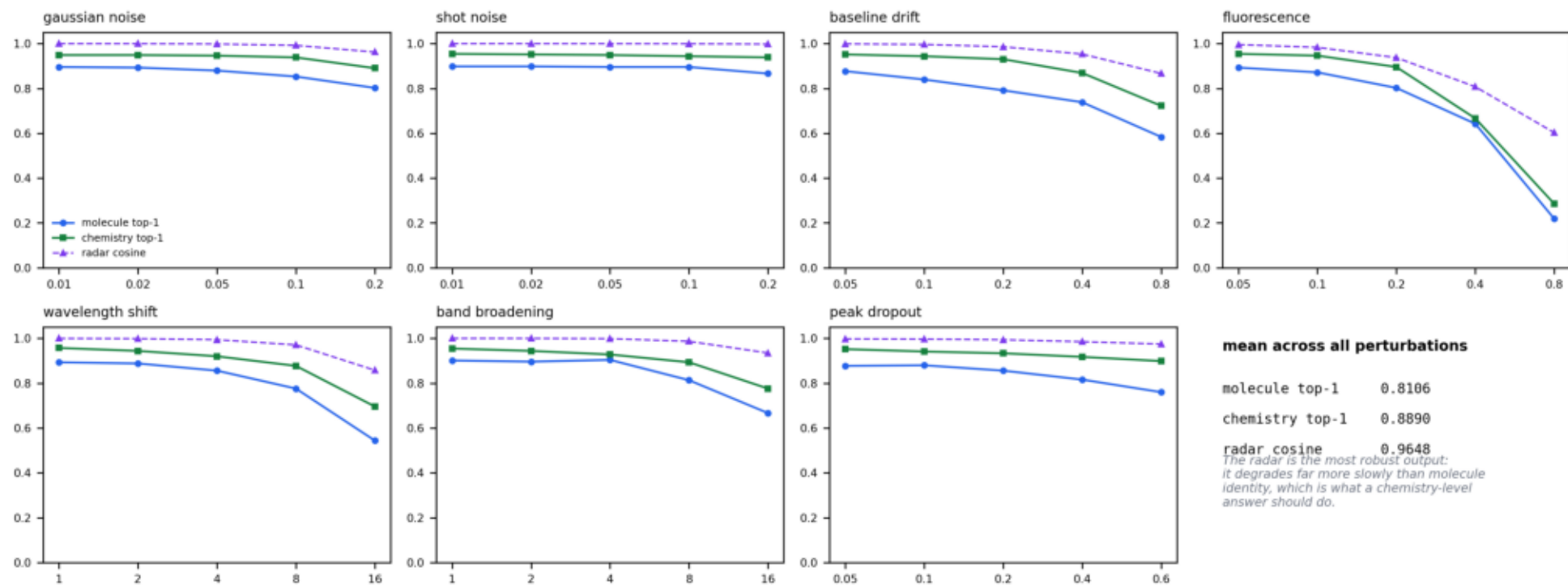
Band broadening and peak dropout hurt molecule identity most, which makes physical sense: both destroy the fine band structure that distinguishes one molecule from its close relative, while leaving the coarse pattern that identifies a family.

In plain terms

Across every condition the ordering holds: radar 0.965 > chemistry 0.889 > molecule 0.811. When evidence weakens, the engine gives up naming the molecule before it gives up describing the chemistry. That ordering is the architecture's central promise, and this figure is where it is demonstrated rather than asserted.

Figure 14

Figure 14 · Noise robustness of the complete engine — 7 perturbations × 5 levels



Noise robustness of the complete engine across seven perturbations at five levels each. The radar degrades far more slowly than molecule identity, which is what a chemistry-level answer should do.

The engine's own weak points

A system that only shows its successes cannot be trusted. These are the failures, by name.

Per-class F1 spans 1.000 down to 0.400. The weak classes are the small ones, and they are weak in a chemically predictable direction.

The low-reconstruction tail is chemically coherent: pyruvate at 0.209, thymine around 0.27, malic acid at 0.26, urea at 0.345. These are small, high-symmetry molecules with few strong bands between 450 and 1800 cm⁻¹. A dictionary built largely from biopolymers should struggle here, and it does - the prediction and the observation agree.

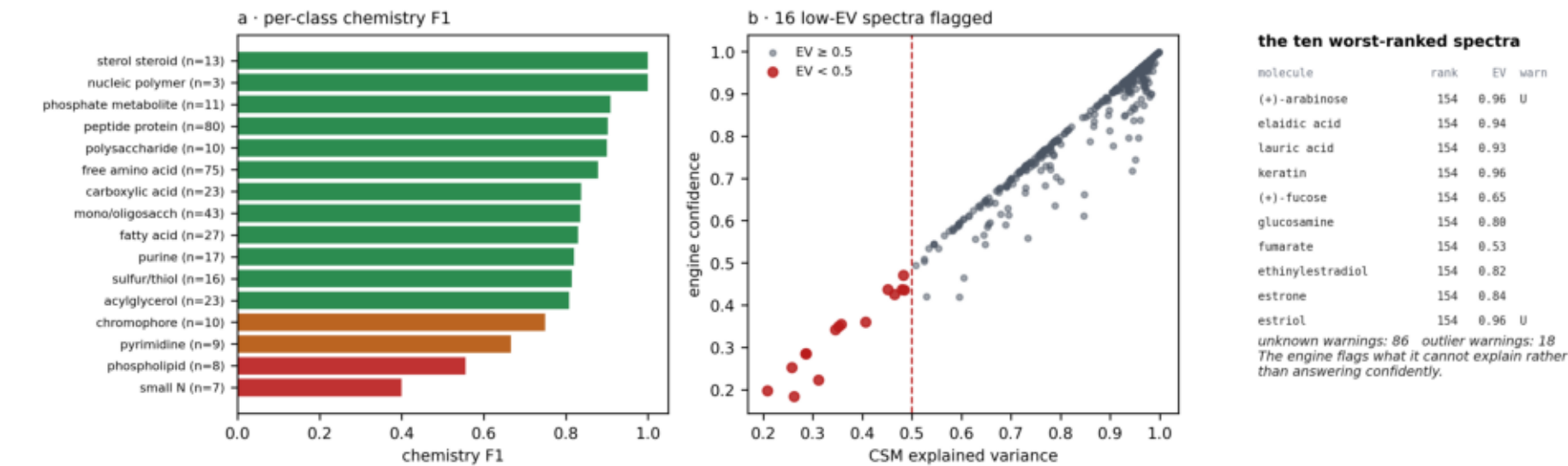
The ten worst-ranked spectra are named individually. Several are singleton molecules whose answer was never in the bank; others, like stearate being called an acylglycerol, are boundary cases where the spectrum is well explained and the ontology is what is difficult.

In plain terms

One limit found while writing the regression tests: white noise is NOT reliably flagged. It reconstructs at explained variance around 0.61, above the 0.50 warning floor, so the flag fires on only 1 of 20 random spectra. Confidence still separates it cleanly - noise peaks at 0.495 against a corpus mean of 0.803. Read the confidence, not the flag.

Figure 15

Figure 15 · Failure analysis



Failure analysis: per-class chemistry F1, the low-reconstruction tail, and the ten worst-ranked spectra by name. The engine flags what it cannot explain rather than answering confidently.

FIGURE 16

The result the architecture rests on

One page, three claims, each measured the same way on the same corpus.

The middle panel is the important one. Chemistry accuracy on molecules the model has never seen rises from 0.608 using the raw spectrum, to 0.850 using the LSM layer, to 0.855 using the CSM layer - and then falls for every layer built on top: 0.664 for eleven grounded axes, 0.405 for themes, 0.392 for Meta Components.

Four independent attempts to build an abstraction above the CSM each lost information. A fifth, the geometric coordinate layer of Phase 06.5, produced an improvement of +0.016 that a paired significance test could not distinguish from zero ($p = 0.180$).

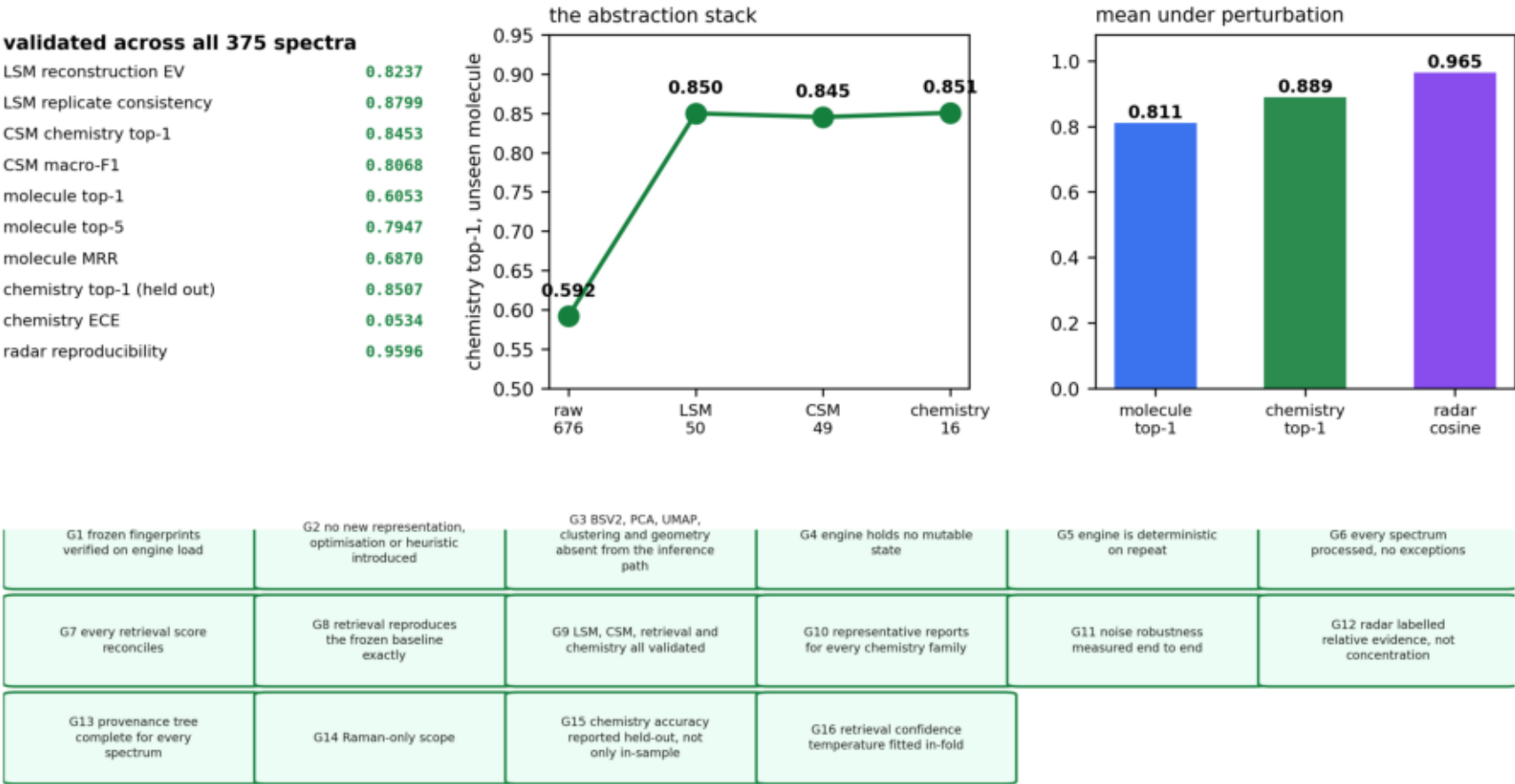
So the engine ships the layer where the information actually is, and treats everything above it - including BSV2 - as a description of that layer rather than a stage of it.

In plain terms

This is not a claim that the CSM representation is optimal. It is the considerably stronger and more useful claim that four separate attempts to improve on it, designed independently and measured identically, all failed.

Figure 16

Figure 16 · End-to-end engine summary



16 of 16 gates pass · atlas 2e43ddcca7d3be41... · deterministic · every score reconciles

End-to-end summary. The middle panel is the finding the whole rebuild rests on: chemistry accuracy on unseen molecules rises from the raw spectrum to the CSM layer and does not rise further.



Results

Four validations across all 375 spectra, the per-class detail behind them, the robustness sweep, and the sixteen gates.

Validations 1 and 2 — representation

Validation 1 — the LSM layer

mean explained variance	0.8237
minimum explained variance	0.2087 (pyruvate)
mean reconstruction error	0.3732
mean active components	9.6 of 50
mean sparsity	0.8723
replicate consistency	0.8799

Ten motifs of fifty reconstruct a typical spectrum, and replicates of the same molecule land in nearly the same place. The dictionary is describing composition, not memorising curves.

Validation 2 — the CSM layer

chemistry-class top-1	0.8453
chemistry-class top-3	0.9707
macro F1	0.8068
balanced accuracy	0.7966
mean explained variance	0.8232
replicate consistency	0.8927

In plain terms

Collapsing 50 motifs to 49 costs almost nothing in reconstruction while raising replicate consistency. That trade is the entire justification for the CSM layer, and it is why the CSM activation is the canonical representation rather than the LSM activation.

Validation 3 — molecular retrieval

Leave-one-spectrum-out over all 375 spectra against the full 154-molecule bank. Every metric reproduces the frozen Phase 05/08 baseline exactly.

metric	Phase 09	frozen baseline
top-1	0.6053	0.6053
top-3	0.7627	0.7627
top-5	0.7947	0.7947
top-10	0.8107	0.8107
MRR	0.6870	0.6870
nDCG@5	0.7112	0.7112
median rank	1.0	—
calibration ECE	0.1205	—
discrimination	0.6914	—

Where the ranks fall

rank	spectra	share
1	227	0.605
2	39	0.104
3	20	0.053
5	12	0.032
10	6	0.016
25	3	0.008
50	0	0.000
100	0	0.000
unretrievable	68	0.181

Note

The final row is structural. 66 of 154 molecules have exactly one spectrum, so when it is held out the correct answer is absent from the bank entirely. These 68 spectra are counted as failures. Excluding them would raise top-1 from 0.605 to roughly 0.74 and would misrepresent what the engine did.

Validation 4 — Chemistry Evidence

Read this first

The shipped engine fits its chemistry map on all 375 spectra, as a shipped engine should. The in-sample figures describe that object. The number to quote as performance is the held-out one.

Held out — molecule-grouped cross-validation

fine-class top-1	0.8507
fine-class top-3	0.9760
macro F1	0.8110

In-sample — a sanity check on the shipped fit

fine-class top-1	0.9547
fine-class top-3	1.0000
macro AUC	0.9990
macro average precision	0.9833

Calibration and stability

expected calibration error	0.0534 (in-sample; held-out was 0.1247 in Phase 06)
Brier score	0.0540
discrimination	0.8529
radar reproducibility	0.9596

In plain terms

The gap between 0.955 and 0.851 is exactly the illusion that in-sample evaluation creates. Phase 09 found this as a defect in its own first draft, fixed it, and added two gates so it cannot recur.

Per-class chemistry behaviour

Per-class behaviour under molecule-grouped cross-validation. All sixteen classes, named explicitly.

chemistry class	n	precision	recall	F1
acylglycerol	23	0.792	0.826	0.809
carboxylic_acid_metabolite	23	0.900	0.783	0.837
chromophore_pigment	10	1.000	0.600	0.750
fatty_acid	27	0.846	0.815	0.830
free_amino_acid	75	0.890	0.867	0.878
mono_oligosaccharide	43	0.792	0.884	0.835
nucleic_acid_polymer	3	1.000	1.000	1.000
peptide_protein	80	0.881	0.925	0.902
phosphate_metabolite	11	0.909	0.909	0.909
phospholipid_sphingolipid	8	0.500	0.625	0.556
polysaccharide	10	0.900	0.900	0.900
purine	17	0.727	0.941	0.821
pyrimidine	9	0.833	0.556	0.667
small_nitrogenous	7	0.375	0.429	0.400
sterol_steroid	13	1.000	1.000	1.000
sulfur_thiol_cofactor	16	1.000	0.688	0.815

Green marks F1 at or above 0.90; amber marks below 0.70. The weak classes are the small ones, and they are weak toward the larger families whose dominant bands they share.

Note

The ontology was not modified in response to this table. Merging two classes because a model confuses them raises the score by removing the distinction the score exists to measure.

Robustness

Seven perturbations, five levels each, applied to the raw spectrum and run through the complete engine. 35 conditions across all 375 spectra.

mean molecule top-1	0.8106
mean chemistry top-1	0.8890
mean radar cosine	0.9648

perturbation	molecule	chemistry	radar	worst radar
band broadening	0.836	0.899	0.984	0.935
baseline drift	0.766	0.884	0.960	0.867
fluorescence	0.686	0.750	0.866	0.603
gaussian noise	0.865	0.935	0.991	0.963
peak dropout	0.838	0.929	0.990	0.975
shot noise	0.891	0.948	0.999	0.998
wavelength shift	0.791	0.879	0.964	0.858

The ordering never reverses. Under every perturbation the radar degrades most slowly, chemistry next, molecule identity fastest.

In plain terms

That is the correct ordering for a chemistry-level answer. As evidence weakens the engine should lose its grip on WHICH molecule before it loses its reading of WHAT KIND of chemistry is present. This table is where that design intention becomes a measurement.

16 of 16 gates pass. Gates are declared before the run and are not relaxed afterwards.

gate	status
G1 frozen fingerprints verified on engine load	PASS
G2 no new representation, optimisation or heuristic introduced	PASS
G3 BSV2, PCA, UMAP, clustering and geometry absent from the inference path	PASS
G4 engine holds no mutable state	PASS
G5 engine is deterministic on repeat	PASS
G6 every spectrum processed, no exceptions	PASS
G7 every retrieval score reconciles	PASS
G8 retrieval reproduces the frozen baseline exactly	PASS
G9 LSM, CSM, retrieval and chemistry all validated	PASS
G10 representative reports for every chemistry family	PASS
G11 noise robustness measured end to end	PASS
G12 radar labelled relative evidence, not concentration	PASS
G13 provenance tree complete for every spectrum	PASS
G14 Raman-only scope	PASS
G15 chemistry accuracy reported held-out, not only in-sample	PASS
G16 retrieval confidence temperature fitted in-fold	PASS

Note

G15 and G16 were added during the phase, after an in-sample Validation 4 and a pinned calibration temperature were found. Adding a gate in response to a defect is permitted; relaxing one in response to a result is not.

IV

Mathematics

Every equation the engine executes, and every metric used to judge it. Each is followed by a plain-English reading of what it actually does.

Preprocessing

Baseline removal — asymmetric least squares

$$z = \operatorname{argmin}_z \sum_b \omega_b (r_b - z_b)^2 + \lambda \sum_b (\Delta^2 z_b)^2$$

$$\omega_b = p \text{ if } r_b > z_b, \quad \omega_b = 1 - p \text{ otherwise}$$

With $\lambda = 1e5$, $p = 0.01$ and ten iterations. The corrected spectrum is $\max(r - z, 0)$.

In plain terms

Two forces are balanced. The first pulls the curve toward the data, but asymmetrically: points above the curve get weight 0.01 and points below get 0.99, so peaks are largely ignored while the valleys between them pull hard. The second term, scaled by the large λ , punishes curvature and forces smoothness. What survives is the slow background, not the sharp bands. A symmetric fit would be dragged up into the peaks and would subtract away real signal.

Normalisation and quality

$$x = \frac{\max(r, 0)}{\|\max(r, 0)\|_2 + \epsilon}, \quad \text{SNR} = \frac{\max_b x_b}{\text{median}_b |x_{b+1} - x_b| + \epsilon}$$

In plain terms

Doubling the laser power doubles every intensity, so scaling to unit length throws that away and keeps only the shape — which is where chemistry lives. It is also why nothing downstream may be read as a concentration. For the noise estimate, using the MEDIAN hop rather than the mean stops real peaks, which produce a few enormous hops, from inflating it.

Learning the dictionaries

The motif dictionaries — non-negative matrix factorisation

$$\min_{W \geq 0, H \geq 0} \|X - WH\|_F^2$$

H holds the motifs, W their per-spectrum weights, and the Frobenius norm is the sum of squared entrywise errors. Learned once in Phase 01, within each chemistry class, and frozen.

In plain terms

This asks: what small set of building-block spectra can be ADDED TOGETHER, never subtracted, to reproduce what was observed? The non-negativity is a scientific constraint. A real mixture spectrum is a sum of its components; there is no such thing as negative-two-parts glucose. Methods that allow negative weights fit better and produce components that mean nothing.

Merging motifs — the consensus graph

$$W_{ij} = \prod_f (\max(F_{ij}^{(f)}, 10^{-3}))^{\alpha_f}, \quad \sum_f \alpha_f = 1$$

Seven similarity features per motif pair: band overlap 0.25, spectral cosine 0.20, peak agreement 0.15, bootstrap co-occurrence 0.15, substitutability 0.10, activation co-occurrence 0.10, provenance overlap 0.05.

In plain terms

A geometric mean is a strict form of 'all of the above'. Under an ordinary average, a spectral cosine of 0.95 could carry an edge whose every other channel was near zero — two motifs that merely look alike would merge. Under a geometric mean, any single near-zero channel drags the whole weight toward zero. The floor at 1e-3 makes the strongest possible veto a stated constant rather than an accident of floating-point representation.

Projection and retrieval

Projection

$$a = \operatorname{argmin}_{a \geq 0} \|x - a^\top C\|_2^2$$

Solved by non-negative least squares against the frozen 49-motif dictionary C. Diagnostics, with the reconstruction $\hat{x} = a^\top C$:

$$EV = 1 - \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2 + \varepsilon}, \quad \rho = \frac{\|x - \hat{x}\|_2}{\|x\|_2 + \varepsilon}$$

In plain terms

'Which mixture of the 49 frozen motifs best explains this spectrum, given that you cannot use a negative amount of anything?'
There is NO regularisation parameter, and that absence is deliberate: a penalty weight would be a quantity that could be tuned per spectrum, and a tunable quantity on the inference path is a place where results can be nudged.

Molecular retrieval

$$S_m = \operatorname{clip}(\hat{R}_m \cdot \hat{q}, 0, 1) = \sum_{j=1}^{49} \hat{q}_j \hat{R}_{mj}$$

In plain terms

Cosine similarity asks whether two activation patterns point in the same DIRECTION, ignoring how large they are. The second equality is why the engine can explain itself: an inner product is literally a sum of per-motif terms, so the score decomposes exactly into 'how much each motif contributed'. The engine asserts that the parts sum to the whole within 1e-9 for every candidate it reports. Nothing is hidden.

Chemistry Evidence

The selected model is D:A_max_idf with lambda 0.5, chosen by nested molecule-grouped cross-validation on inner-fold macro F1.

Fine level

$$e_c = w_c \cdot \max_{m \in M_c} S_m, \quad w_c = \frac{\log(N/|M_c|) + 1}{\text{mean}_{c'}[\log(N/|M_{c'}|) + 1]}$$

In plain terms

A class's evidence is the similarity of its SINGLE BEST-MATCHING member, not an average — averaging would penalise a chemically diverse class such as free amino acids for containing members unlike the query, which is not a defect. The inverse-frequency weight offsets the opposite bias: with 80 peptide references and 3 nucleic-acid-polymer references, the large class would otherwise win simply by having more chances.

Broad level, soft routing

$$e_c \leftarrow e_c \cdot (B_{\beta(c)})^\lambda, \quad \lambda = 0.5$$

In plain terms

Coarse chemistry is easier to get right than fine chemistry, so it is used as a HINT — but only as a hint. Because every broad evidence is strictly positive, multiplying can never zero a fine class out; a molecule whose superclass was misjudged is penalised, not excluded. A hard filter would make every coarse error permanently unrecoverable.

The radar

$$\tilde{e} = \frac{e}{\sum_c e_c + \epsilon}$$

Note

This quantity is RELATIVE BIOCHEMICAL EVIDENCE. Not a concentration, not an abundance, not a mixture fraction. Normalisation destroyed absolute scale in the first stage, and a similarity is not a quantity of material.

Calibration

Temperature scaling

$$p_c = \frac{\exp(e_c/T)}{\sum_{c'} \exp(e_{c'}/T)}, \quad T = 0.4538$$

In plain terms

One number rescales how sharply evidence becomes probability. It cannot change the RANKING — the winner is the winner at any T — only how confident the engine claims to be. That is exactly what is wanted from a calibrator: fix the honesty of the numbers without touching the answer.

Expected calibration error

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|$$

In plain terms

Of everything the engine called 80% likely, was it right 80% of the time? ECE averages that gap. But ECE ALONE IS DANGEROUS: a model that outputs the base rate for every input has near-perfect ECE and is useless. V7 hit this exact trap in Phase 05. Selection therefore uses log loss or Brier subject to floors on sharpness and discrimination, so a constant predictor is disqualified before ECE is ever consulted.

Sharpness, discrimination, risk-coverage

$$\text{sharpness} = \text{Var}_i(p_i, \hat{y}), \quad \text{disc} = \Pr(p^{\text{correct}} > p^{\text{wrong}})$$

At full coverage retrieval accuracy is 0.605. Abstaining below a margin of 0.497 keeps 51% of spectra at 0.791.

In plain terms

Sharpness asks whether the confidences vary at all; a constant predictor scores zero. Discrimination asks whether the engine is more confident when it is right than when it is wrong. The risk-coverage curve turns both into an operational rule: answer only above this threshold, and here is what you get.

Metrics and uncertainty

Retrieval metrics

$$\text{top}_k = \frac{1}{n} \sum_i \mathbb{1}[\text{rank}_i \leq k], \quad \text{MRR} = \frac{1}{n} \sum_i \frac{1}{\text{rank}_i}$$

In plain terms

Top-k is the blunt question — was the answer in the first k? MRR rewards being close: rank 1 scores 1.0, rank 2 scores 0.5, rank 10 scores 0.1, so a system that is usually second beats one that is usually tenth.

Classification metrics

$$\text{macro F1} = \frac{1}{16} \sum_c \frac{2P_c R_c}{P_c + R_c}, \quad \text{balanced acc} = \frac{1}{16} \sum_c R_c$$

In plain terms

Plain accuracy is dominated by the big classes — with 80 peptide spectra and 3 nucleic-acid-polymer spectra, a model could ignore the small classes entirely and still look good. Macro F1 gives every class one vote. That is why macro F1 (0.811) sits below top-1 (0.851): the engine is worse on the rare classes, and macro F1 refuses to let that hide.

Uncertainty

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

In plain terms

McNemar's test. When two systems run on the SAME spectra, only the cases where they disagree carry information — the ones both get right tell you nothing about which is better. This test is the reason Phase 06.5 did not adopt the geometry layer: the raw improvement of +0.016 came from six spectra, and McNemar returned $p = 0.180$.

Bootstrap confidence intervals resample MOLECULES, not spectra. Replicate spectra of glucose are not independent evidence about how well the engine handles sugars, and treating them as independent would shrink the interval to something the data does not support.



Limits, decision and glossary

What this engine may and may not be used to claim, what the decision gate concluded, and the vocabulary used throughout.

Interpretation limits

These are constraints on what the output MEANS, not caveats about its quality.

- The radar is relative biochemical evidence. Not a concentration, not an abundance, not a mixture decomposition.
- Pure Raman reference spectra only. Nothing here licenses a SERS, serum, plasma, EV, tissue or pathogen claim. Transfer to those regimes is unmeasured in V7 and must be established separately.
- A peak is not a molecule. Retrieval returns ranked candidates with a score decomposition, deliberately, and top-1 of 0.605 is the honest figure for how often the single best guess is right.
- The sixteen classes are a curated cut through a continuum. Phase 06.5 showed the space has NO preferred cluster count; the ontology is a reporting convention that generalises well, not a natural kind.
- In-sample chemistry figures describe the shipped fit, not expected performance on new molecules.
- The engine cannot tell that the true molecule is absent from its bank. The 'unknown' and 'outlier' warnings detect unexplained SPECTRA, not unknown MOLECULES. Using them as an open-set detector would be a misuse.
- Class-prior bias remains open. Class sizes range from 3 to 80 spectra and the chemistry layer inherits that prior; the inverse-frequency weight mitigates it without eliminating it.

Note

Every number in this document comes from the same 375 spectra that built the atlas. The robustness study is the closest thing to an external test and it is synthetic. The correct reading of Phase 09 is 'the architecture is internally sound and faithfully implemented' — not 'the architecture works'. The second claim needs data V7 does not have.

The decision

Phase 09 was declared a packaging phase before any code ran. Its gate asks whether the frozen architecture is faithfully implemented and honestly validated — not whether some change is worth adopting.

Outcome A — ship the engine and freeze V7

Retrieval reproduces the Phase 05/08 baseline to the digit. The chemistry layer, the CSM projection and the LSM projection all reproduce their source phases. Integrating eight phases into one object surfaced no contradiction between them, which is the strongest single piece of evidence that the chain of decisions behind the rebuild is sound.

What is now frozen

Canonical preprocessing; the 50-motif and 49-motif dictionaries; retrieval by CSM cosine; the Chemistry Evidence model and its calibrator; the confidence and warning rules; the engine's API. Changing any of them changes a fingerprint and the engine refuses to load — the freeze is enforced by code, not convention.

What is not frozen

BSV2, which sits downstream and may be revised freely. Presentation choices, including where an operator sets an abstention point. And the corpus, whose expansion is the intended path forward.

Where the work goes next

- Corpus. 66 of 154 molecules have a single spectrum, which alone caps molecule top-1 at 0.819. This is the largest single lever available and it needs no engine change.
- Transfer. Establish what survives the move from pure Raman to SERS, serum and EV — the question GAIRA ultimately exists to answer.
- Open-set behaviour. Hold out whole molecules from the bank and measure whether confidence drops. This is currently an argued limitation rather than a measured one, and it is the limitation most likely to matter in use.

Recommendation

Four independent attempts to add a layer above the CSM have now failed on measurement. A fifth would be a re-run of the same experiment under a different name. The productive next work is scientific, not architectural.

Glossary

Raman spectrum	the pattern of light-frequency shifts produced by molecular vibrations in a sample
cm-1 (wavenumber)	the unit of that shift; this engine uses 450 to 1800
asLS	asymmetric least squares — fits the fluorescence background
Savitzky-Golay	a smoother that fits small polynomials, preserving peaks
L2 normalisation	scaling a spectrum to unit length, discarding intensity
NMF	non-negative matrix factorisation — finds add-only building blocks
LSM	Local Spectral Motif — one of 50 basis spectra learned per class
CSM	Consensus Spectral Motif — one of 49 merged motifs; the canonical coordinates
activation	how much of each motif a spectrum contains; never negative
explained variance	the fraction of a spectrum the dictionary accounts for
cosine similarity	how closely two patterns point in the same direction
Chemistry Evidence	the 16 numbers, one per chemistry family, that the radar displays
BSV2	a compressed description of Chemistry Evidence; NOT on the inference path
calibration	making stated confidence match observed correctness
ECE	expected calibration error — the average gap between the two
held out	evaluated on molecules the model never saw during fitting
in-sample	evaluated on the same data used for fitting; not a performance claim
macro F1	an average over classes that gives small classes equal weight
top-k	the fraction of queries whose answer appeared in the first k results
MRR	mean reciprocal rank — rewards being close, not just being first
fingerprint	a hash that changes if a frozen artefact changes at all
gate	a pass/fail condition declared before a run and never relaxed after

Colophon and provenance

Every figure, table and number in this document was generated from the committed artifacts of Phase 09 by the scripts listed below. No value was typed by hand.

engine	src/gaira/v7/canonical/engine.py
validation	results/v7_rebuild/phase09/code/run_phase09.py
figures	results/v7_rebuild/phase09/code/make_figures.py
this document	results/v7_rebuild/phase09/code/make_pdf.py
tests	tests/test_v7_phase09.py
atlas fingerprint	2e43ddcca7d3be41c5f9da016fb8277f
frozen atlas	09ed804a40836f4a05a91ba10900cded
frozen LSM registry	208482d6f7178b5b8f16cace91be55b0
frozen CSM registry	0b4aa550ccefed3edabdbde5bae11c8d
frozen Phase 05 engine	20d8bd99ce71f45a125c6a2b1d719e51
completed	2026-08-07 05:32:15 UTC

Companion documents

- PHASE_09_REPORT.md — the findings
- PHASE_09_ENGINE_SPEC.md — the engine, precisely enough to reimplement
- PHASE_09_MATHEMATICAL_APPENDIX.md — every equation, with intuition
- PHASE_09_DECISION_GATE.md — the criteria and the decision
- PHASE_09_SCIENTIFIC_AUDIT.md — what is strongly supported, weakly supported, and unsupported